Repeated-Sprint Training at 5000-m Simulated Altitude in Preparation for the World Rugby Women's Sevens Series: Too High?

FRANCK BROCHERIE[1], SEBASTIEN RACINAIS[2], SCOTT COCKING[3], NATHAN TOWNSEND[2], ANTHONY COUDERC[4], JULIEN PISCIONE[4], and OLIVIER GIRARD[2,5]

[1]Laboratory Sport, Expertise and Performance (EA 7370), French Institute of Sport (INSEP), Paris, FRANCE; [2]Research and Scientific Support, Aspetar Orthopaedic and Sports Medicine Hospital, Doha, QATAR; [3]Aspire Academy, Doha, QATAR; and [4]Research Department, French Rugby Union Federation (FFR), Marcoussis, FRANCE; and [5]School of Human Science (Exercise and Sport Science), The University of Western Australia, Perth, AUSTRALIA

ABSTRACT

BROCHERIE, F., S. RACINAIS, S. COCKING, N. TOWNSEND, A. COUDERC, J. PISCIONE, and O. GIRARD. Repeated-Sprint Training at 5000-m Simulated Altitude in Preparation for the World Rugby Women's Sevens Series: Too High? *Med. Sci. Sports Exerc.*, Vol. 55, No. 10, pp. 1923–1932, 2023. **Purpose:** The objective of this study is to investigate the effectiveness of novel repeated-sprint training in hypoxia (RSH) protocol, likely maximizing hypoxic stimulus (higher than commonly used) while preserving training quality (interset rest in normoxia). **Methods:** Twenty-three world-class female rugby sevens players performed four repeated-sprint training sessions (4 sets of 5 × 5-s cycle sprints—25-s intersprint recovery and 3-min interset rest) under normobaric hypoxia (RSH, exercise and interset rest at FiO_2 of 10.6% and 20.9%, respectively; n = 12) or normoxia (repeated-sprint training in normoxia; exercise and interset rest at FiO_2 of 20.9%; n = 11) during a 9-d training camp before international competition. Repeated-sprint ability (8 × 5-s treadmill sprints—25-s recovery), on-field aerobic capacity, and brachial endothelial function were assessed pre- and postintervention. **Results:** Arterial oxygen saturation (pooled data: 87.0% ± 3.1% vs 96.7% ± 2.9%, P < 0.001) and peak and mean power outputs (sets 1 to 4 average decrease: −21.7% ± 7.2% vs −12.0% ± 3.8% and −24.9% ± 8.1% vs −14.9% ± 3.5%; both P < 0.001) were lower in RSH versus repeated-sprint training in normoxia. The cumulated repeated-sprint distance covered significantly increased from pre- to postintervention (+1.9% ± 3.0%, P = 0.019), irrespective of the condition (P = 0.149). On-field aerobic capacity did not change (all P > 0.45). There was no significant interaction (all P > 0.240) or condition main effect (all P > 0.074) for any brachial artery endothelial function variable. Only peak diameter increased (P = 0.026), whereas baseline and peak shear stress decreased (P = 0.014 and 0.019, respectively), from pre- to postintervention. **Conclusions:** In world-class female rugby sevens players, only four additional repeated-sprint sessions before competition improve repeated-sprint ability and brachial endothelial function. However, adding severe hypoxic stress during sets of repeated sprints only did not provide supplementary benefits. **Key Words:** BRACHIAL ENDOTHELIAL FUNCTION, NORMOBARIC HYPOXIA, ALTITUDE TRAINING, REPEATED-SPRINT ABILITY, TEAM SPORTS

Repeated-sprint training in hypoxia (RSH) is a popular "live low–train high" altitude/hypoxic training intervention involving the repetition of short (<30 s) "all-out" efforts with incomplete recoveries (<60 s) in oxygen-deprived environments (1–3). A 2017 meta-analysis including nine studies (all published between 2013 and 2016) concluded that RSH boosts repeated-sprint ability and sport-specific physical performance compared with similar training in normoxia (repeated-sprint training in normoxia (RSN)) (1). In recent years, the interest for RSH has grown steadily with more than 25 RSH studies now available (4) since the pioneering experiment in 2013 (2). Although not a universal finding (5,6), putative RSH benefits have been evidenced for a range of team (e.g., soccer, rugby union, and field hockey), racquet (e.g., tennis), and endurance sports (e.g., cycling and cross-country skiing) (1,2,7–12).

A large number of previous RSH studies conducted in normobaric hypoxia have recruited male rugby union and rugby league players (7,13,14). In well-trained academy rugby union and rugby league players, Galvin et al. (13) reported twofold greater gains for intermittent running performance after 12 RSH (i.e., 10 × 6-s sprints—30-s passive recovery, sessions performed at 3500 m) versus RSN sessions over 4 wk. Shorter RSH intervention periods using real-world scenarios (i.e., four and six sessions over two "in-season" and three "preseason" weeks, respectively) also resulted in greater improvements compared with RSN in world-class (7) and well-trained (14) rugby union male players. To date, the only study involving

Address for correspondence: Franck Brocherie, Ph.D., Laboratory Sport, Expertise and Performance (EA 7370), French Institute of Sport (INSEP), 11 Avenue de Tremblay, 75012 Paris, France; E-mail franck.brocherie@insep.fr.
Submitted for publication January 2023.
Accepted for publication May 2023.

0195-9131/23/5510-1923/0
MEDICINE & SCIENCE IN SPORTS & EXERCISE®

DOI: 10.1249/MSS.0000000000003226

team-sport (lacrosse) college female athletes demonstrated two- to threefold larger increases in sprinting peak and mean power outputs (peak power output [PPO] and MPO, respectively) after eight RSH sessions (i.e., 2 × 10 × 7-s sprints—30-s passive recovery, sessions performed at 3000 m) performed over 4 wk (15). With females being less sensitive to hypoxic stimuli (16) and more fatigue resistant than males during repeated sprints (17), the efficacy of RSH in (elite or world class) female athletes is largely unknown.

For optimal implementation, RSH requires sport-specific adjustments of the main training variables, in turn modifying the contribution of glycolysis to energy supply and recruitment of fast-twitch fibers (1). This can be achieved by altering the length/duration of sprint and recovery intervals, exercise/recovery ratio, interset rest duration, session frequency, as well as "hypoxic dose" (hypoxia severity and/or duration). A careful examination of available RSH studies (1) indicates that a rather narrow range (2900 to 3500 m) of simulated altitude has typically been selected, yet with no strong physiological rationale for implementing moderate hypoxia levels. Perhaps simulated altitudes above 3500 m were not selected to avoid blunting absolute training quality due to well-described earlier and larger performance decrements at higher elevations (18,19). Yet, in severe acute hypoxia (~5300 m), PPO and total work performed on a single Wingate test were maintained despite reduced aerobic energetic contribution (20). Weyand et al. (21) also reported no significant effect of hypoxia (~3800 m) on "all-out" sprint performance over durations up to ~60 s, alongside an increased oxygen deficit. Conversely, over longer total duration (longer than 30 to 60 s), repeated-sprint ability (8 × 5-s treadmill sprints—25-s intersprint recovery) was impaired at a simulated altitude of ~3600 m compared with both ~1800 m and normoxia (22). Despite reduced external workload in the most severe hypoxic condition (~3600 m), larger peripheral fatigue (i.e., twitch force amplitude of knee extensors after femoral nerve stimulation) was also incurred after repeated treadmill sprints (23). Collectively, these studies suggest that performing "all-out" efforts in very severe hypoxia (~5000 m) may activate anaerobic and neuromuscular pathways beyond that which can be achieved at sea level or moderate hypoxia (~3000 m). Because a longer total duration of continuous exposure to extreme hypoxia—e.g., during an entire repeated sprint session—could also lead to a decrease in training quality, observing interset rest in normoxia may be required.

Suggested RSH adaptive mechanisms for performance enhancement include improved muscle blood perfusion (11), optimized oxygen extraction by fast-twitch fibers (2), and specific skeletal muscle adaptations (molecular level) that may arise through oxygen-sensing pathways (i.e., hypoxic-inducible factors) (2,10). To maintain oxygen delivery to tissues, hypoxic-induced vasoconstriction (24) is compensated by a nitric oxide-dependent increase in blood flow and shear stress when exercising in hypoxia (25), which reflects an improved endothelial function. Using an animal model, recent evidence indicates that RSH further enhances vascular reactivity (i.e., increases in both acetylcholine-induced vasorelaxation (+20%) and phenylephrine-induced vasoconstriction (+120%)) compared with RSN (26). However, whether vascular function assessed via the flow-mediated dilation (FMD) technique (27) is improved in humans after RSH is unsubstantiated.

Therefore, the aim of this study was to investigate the effectiveness of a novel RSH protocol to boost performance and physiological adaptations in world-class female rugby sevens players using a realistic setting (i.e., four additional RSH sessions over a 9-d precompetition training camp). Our intention was to prescribe a very severe hypoxic stimulus (FiO_2 = 10.6%, equivalent to ~5000-m simulated altitude) during sets of repeated sprints to maximize (anaerobic) training stimulus while observing interset rest in normoxia to facilitate recovery and preserve training quality. We hypothesized that sport-specific physical performance gains (repeated-sprint ability and on-field aerobic capacity) and vascular function adaptations will be larger after this innovative RSH protocol compared with RSN.

METHODS

Subjects

Twenty-four world-class female players (24.2 ± 3.6 yr, 171.7 ± 7.3 cm and 69.0 ± 6.8 kg) belonging to the French national rugby sevens team participated in the study as part of their normal squad training schedule in preparation for a World Rugby Women's Sevens Series tournament. Subjects gave their written informed consent after being informed of all experimental procedures (e.g., severe intensity nature of the proposed exercise and hypoxic stimulus), and possible risks (e.g., headache, dizziness, tiredness, shortness of breath, and nausea, in isolation or in combination), associated with the experiment. Exclusion criteria for participation were exposure to hypoxia >2000 m (FiO_2 of 16.1%) for more than 48 h during a period of 6 months before the study. During the study, one subject was excluded because of injury. The experiment was approved by the *Anti-Doping Laboratory Ethics Committee* in Qatar (institutional review board approval E2017000254) and conformed to current *Declaration of Helsinki* guidelines.

Study Design

This randomized, single-blinded, placebo-controlled study consisted of two testing sessions before (pre-) and immediately after (post-) a 9-d supervised specific training period (Fig. 1). According to their initial fitness level (primary factor) and playing position (secondary factor), subjects were assigned to one of the two experimental groups with repeated-sprint training either in normobaric hypoxia (RSH, n = 12) or normoxia (RSN, n = 11) (Table 1). In addition to their regular training/recovery routine, both groups undertook four supplementary RSH/RSN training sessions over the 9-d intervention period starting the day after pretests and finishing 48 h before posttests. The time of day for training and testing was consistent both within and between groups. Pre- and posttesting sessions were performed in normoxia over two consecutive days in the

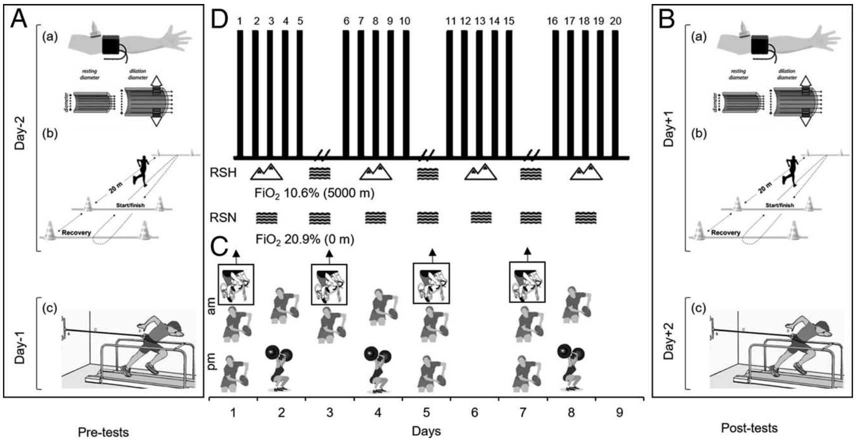

FIGURE 1—Experimental overview. Pre- (A) and posttests (B) include assessment of FMD (a), the distance covered during YYIR2 (b), and treadmill repeated-sprint ability (c). Training procedure (C) featuring four repeated-sprint training sessions (i.e., four sets of 5 × 5-s cycle ergometer sprints—25-s intersprint recovery and 3-min interset rest) under normobaric hypoxia (RSH; exercise at FiO_2 of 10.6% corresponding to a simulated altitude of ~5000 m with interset rest in normoxia; *n* = 12) or normoxia (RSN, exercise and interset rest in normoxia; *n* = 11) within a 9-d intervention period (D) also including rugby (*n* = 10) and resistance training (*n* = 3) sessions.

same sequence as follows: (i) day 1, FMD and on-field aerobic capacity (Yo-Yo intermittent recovery level 2 (YYIR2)); (ii) day 2, repeated-sprint ability. Because of the exhaustive nature of the tests, testing sessions were performed after light training in the preceding 24 h and at least 3 h after a meal. In all cases, the team's nutritionist controlled and standardized meals during the intervention, with no alcohol or caffeine consumption permitted for 24 h before testing sessions. Bottled water was provided *ad libitum*. For all tests, subjects were vigorously encouraged during all efforts. Before each test, a specific standardized 15-min warm-up was supervised by two investigators. The medical team was aware of the players' menstrual cycle, but it remained confidential.

Supervised Training Protocol

In order to limit additional impact forces induced by maximal running velocity, which may increase the risk of lower-limb injuries, and to ensure subjects' adherence, repeated-sprint training was performed on the same air-braked cycle ergometer (Wattbike Ltd., Nottingham, United Kingdom). Saddle and handlebar height and position were replicated for every session for each subject. The gearing was self-selected by subjects during the warm-up of the first session and then replicated during all training sessions. Sessions were conducted in thermo-neutral conditions (air temperature, 24°C; relative humidity, 40%) in a normobaric hypoxic chamber (LOXY^{MED} room system; L.O.T. Low Oxygen Technology GmbH, Berlin, Germany) at the same time of the day (±1 h) and 36 h apart. For RSH, FiO_2 was set to 10.6% to simulate an altitude of ~5000 m. To blind subjects, the system was also run for RSN with a normoxic airflow (FiO_2 = 20.9%) into the chamber.

Each session lasted ~50 min, including a ~ 15-min standardized warm-up in normoxia before entering the normobaric hypoxic chamber for the repeated-sprint training routine (~25 min), ending with a ~10-min recovery phase in normoxia. Specifically, the repeated-sprint training routine included four sets of 5 × 5-s cycle sprints—25-s intersprint recovery performed in hypoxia (RSH) or in normoxia (RSN) with 3-min interset rest in normoxia for both groups (exit and entry from and to the chamber within 30 s). Subjects were given a countdown and encouraged to complete every sprint maximally. For each repetition, PPO and MPO (calculated over the entire 5-s effort) were recorded by the cycle ergometer software. Physiological responses (i.e., heart rate (HR), peripheral capillary oxygen saturation (SpO_2) via a wireless Polar monitoring system (Polar Electro Oy, Kempele, Finland) and noninvasive pulse oximeter using a finger probe (GO_2™ Achieve 9570-A; Nonin, Plymouth, MN), respectively) and ratings of perceived exertion (RPE) (6 to 20 Borg's scale) were measured after each training session (28). Training loads were calculated as total training duration (min) × RPE.

Rugby Sevens and Strength and Conditioning Sessions

On average, subjects practiced ~11 to 13 h·wk^{-1} (i.e., 10 rugby sevens sessions, 3 strength and conditioning (including strength-based and/or power-oriented exercises) sessions, and 5 recovery (cold water immersion) sessions) during the 9-d intervention. All rugby sevens training sessions were monitored using a 16-Hz global positioning system (sensorEverywhere; Digital Simulation, France). Derived variables included the total distance covered, averaged relative distance, high- and very

TABLE 1. Subject characteristics.

	RSH (*n* = 12)	RSN (*n* = 11)
Age (yr)	25.6 ± 4.0*	22.6 ± 2.4
Height (cm)	171.7 ± 8.6	171.8 ± 6.1
Weight (kg)	68.7 ± 8.8	69.3 ± 3.9
30-m sprint time (s)	4.51 ± 0.14	4.51 ± 0.15
MAV (km·h^{-1})	15.4 ± 1.2	15.4 ± 1.0
Position	1 FW/4 BW/7 FW–BW	2 FW/1 BW/8 FW–BW

Values are presented as mean ± SD.
* $P < 0.05$, significant difference between groups.
BW, backward; FW, forward; FW–BW, forward–backward; MAV, maximal aerobic velocity.

high-intensity distance covered (i.e., defined as the distance covered at velocity above individual maximal aerobic velocity and above 85% of maximal sprint velocity) (29).

Blinding

Experiment was run in a single-blind manner where subjects in both RSH and RSN groups were told (based on the head coach's request to increase team motivation) that they were actually training under the hypoxic condition (without any specific information about severity of the hypoxic stimuli inside the chamber). For motivational reasons and reinforcement of the subjects' blinding to group classification, they were assigned to different teammates during each specific training session.

Performance Tests

Repeated-sprint ability. The repeated-sprint ability test was performed on an instrumented motorized treadmill (ADAL3D-WR; Medical Development, France) and consisted of performing eight 5-s maximal sprints interspersed with 25 s of passive recovery. Briefly, this motorized treadmill allows subjects to complete accelerations from a still position and to reach high running velocities due to the use of a constant motor torque mode. After a treadmill familiarization procedure followed by a 15-min standardized warm-up (30), a single 5-s sprint was performed. This sprint was used as a criterion score for the subsequent repeated sprints to ascertain that no pacing occurred. Subjects were instructed to perform all sprints maximally, although no specific information about performance was provided after each effort. Repeated-sprint ability was assessed from the distance covered and velocity data using five scores: the largest (i.e., during the first sprint in all cases) distance ran and maximal velocity (V_{max}, first sprint), the cumulated distance covered over the eight sprints (i.e., sum of the eight sprints) and mean velocity (V_{mean}, average of eight sprints), and the sprint decrement score (S_{dec}) [i.e., ([cumulated distance/(largest distance × 8)] − 1) × 100] (30).

On-field aerobic capacity. The YYIR2 was conducted to assess high-intensity intermittent running performance (31). Briefly, this incremental running test to exhaustion involves repeated 20-m shuttle runs at progressively increasing velocity (starting at 13 km·h^{-1}) controlled by audio beeps interspersed with 10-s active recovery periods. The test was ended when the subjects failed to reach the finishing line on the audio beep for two consecutive shuttles. The YYIR2 total distance covered was recorded. This test is reproducible and is a sensitive tool for assessing aerobic capacity in team-sport players (31).

Brachial Artery Endothelial Function

Brachial artery endothelium-dependent function was evaluated using the FMD technique (27). Measurements were performed in the supine position. After 5 min of rest, subjects' arm was extended and positioned at an angle of ~80° from the torso to examine brachial artery FMD. A cuff attached to a rapid inflator (Hokanson, Bellevue, WA) was placed distal to the olecranon process on the forearm to provide a stimulus to ischemia. A 15-MHz multifrequency linear array probe attached to a high-resolution ultrasound machine (T3200T; Terason, Burlington, MA) was used to image the brachial artery in the distal third of the upper arm. When a suitable image was obtained, the probe was held in position and ultrasound settings were altered to optimize the longitudinal B-mode image of the lumen–arterial wall interface. Continuous Doppler velocity was measured using the lowest isonation angle (<60°). After 1 min of baseline recording, the forearm cuff was inflated (>200 mm Hg) for 5 min. Lumen diameters (i.e., baseline diameter and peak diameter defined as the time from cuff release to peak diameter response) and shear rate area under the curve (SR_{AUC}, defined as the measured shear from the point of cuff release to peak dilation) recordings resumed 30 s before cuff deflation and continued for 3 min thereafter, in accordance with current recommendations (32).

Commercial edge detection software (FMD Studio, Quipu, Italy) was used to analyze all recordings independently of investigator bias. As previously described (32,33), blood flow (the product of lumen cross-sectional area and Doppler velocity) was determined from the synchronized diameter and velocity data at a sampling rate of 30 Hz. Shear rate (s^{-1}) (independent of viscosity) was calculated as four times mean blood velocity/vessel diameter. All files were analyzed by the same investigator to maximize reliability between trials (previous coefficient of variation in baseline arterial diameter measurements from our laboratory = 2.2% ± 1.4%). Baseline diameter was also controlled before the introduction of hyperemia in each FMD test to better scale the changes in diameter compared with simple percentage change, which makes implicit assumptions about the relationship between baseline diameter and peak diameter (34).

Statistical Analysis

Values are expressed as mean ± SD or relative changes (%) unless otherwise stated. Normal distribution of the data was tested using the Shapiro–Wilk test. Differences in training response between groups were tested using one-way analysis of variance (ANOVA) and two-way repeated measures ANOVA (condition × exercise time points or repetitions). Test performance, physiological, and vascular function variables were compared using two-way repeated measures ANOVA with one between factor (condition, RSH vs RSN) and one within factor (time, pre- vs post-). Multiple comparisons were made using Holm–Sidak *post hoc* test. Effect sizes were described in terms of partial eta-squared (η_p^2, with $\eta_p^2 \geq 0.06$ representing a moderate effect and $\eta_p^2 \geq 0.14$ a large effect). All analyses were made using Sigmaplot 11.0 software (Systat Software, Inc., San Jose, CA). Null hypothesis was rejected at $P < 0.05$.

RESULTS

Training Response

Supervised training protocol. Significant condition–time interaction ($P < 0.001$, $\eta_p^2 = 0.15$), condition ($P < 0.001$, $\eta_p^2 = 0.19$), and time main effects ($P < 0.001$, $\eta_p^2 = 0.27$) were found for SpO_2 (pooled over the four sessions). *Post hoc* analysis revealed significant differences between RSH and RSN during the exposure in the normobaric hypoxic chamber (preexercise: 88.0% ± 3.0% vs 96.9% ± 2.5%; postexercise: 76.8% ± 6.7% vs 95.0% ± 6.9%; both $P < 0.001$, $\eta_p^2 > 0.64$) but not during the normoxic between-sets rest period (96.5% ± 1.2% vs 97.8% ± 0.7%; $P = 0.143$, $\eta_p^2 = 0.30$) (Fig. 2A). During the exercise sets, HR (pooled over the four sessions) followed a similar pattern over time ($P < 0.001$, $\eta_p^2 = 0.82$), with no difference between groups ($P = 0.120$, $\eta_p^2 = 0.00$) and no significant condition–time interaction ($P = 0.695$, $\eta_p^2 = 0.00$) (Fig. 2B). Significant condition (all $P < 0.001$, $\eta_p^2 > 0.02$) and time (all $P < 0.001$, $\eta_p^2 > 0.07$) main effects were found for averaged PPO and MPO across the four sessions. During the first repetition of each set, PPO and MPO did not differ between conditions (all $P > 0.375$). Sets 1 to 4 average decrease in PPO (−21.7% ± 7.2% vs −12.0% ± 3.8%; $P < 0.001$, $\eta_p^2 = 0.42$; Figure 2C) and MPO (−24.9% ± 8.1% vs −14.9% ± 3.5%; $P < 0.001$, $\eta_p^2 = 0.39$) (Fig. 2D) was larger for RSH than RSN. There was no significant difference in RPE ($P = 0.064$, $\eta_p^2 = 0.09$) monitored after both RSH (16 ± 1 a.u.) and RSN (15 ± 1 a.u.) sessions. Similarly, there were no significant differences between conditions for mean training load (RSH, 785 ± 55, vs RSN, 747 ± 3 a.u.; $P = 0.063$, $\eta_p^2 = 0.09$) or total training load (RSH, 2986 ± 55, vs RSN, 2989 ± 121 a.u.; $P = 0.531$, $\eta_p^2 = 0.00$).

Rugby sevens and strength and conditioning sessions. Regarding rugby sevens training load, training duration (16.0 ± 0.5 vs 16.0 ± 0.6 h; $P = 0.980$, $\eta_p^2 = 0.01$), the total distance covered (51.5 ± 1.9 vs 50.5 ± 1.8 km; $P = 0.860$, $\eta_p^2 = 0.06$), averaged relative distance (56.5 ± 27.1 vs 55.5 ± 26.7 m·min^{-1}; $P = 0.800$, $\eta_p^2 = 0.01$), and averaged percentage of the high-intensity distance covered (13.5% ± 7.9% vs 15.1% ± 9.1%; $P = 0.640$, $\eta_p^2 = 0.01$) did not differ between RSH and RSN. Although not strictly controlled, all efforts were made to match strength and conditioning content, volume, and intensity (i.e., sessions including two strength-based exercises (three to four sets of one to four repetitions at ~90% 1 repetition maximum) and two power-oriented exercises (four to five sets of three to six repetitions at ~60% 1 repetition maximum)) between groups.

Exercise Performance Tests

Repeated-sprint ability. The distance covered and associated running velocity during the repeated-sprint exercise test are displayed in Figure 3. There was no significant condition–time interaction ($P = 0.487$, $\eta_p^2 = 0.00$) or any condition ($P = 0.451$, $\eta_p^2 = 0.11$) or time main effects ($P = 0.130$, $\eta_p^2 = 0.00$) for the distance covered during the first sprint (RSH: 25.3 ± 1.3 vs 25.6 ± 1.2 m from pre- to post-; RSN: 24.5 ± 1.4 vs 24.6 ± 1.6 m from pre- to post-) (Fig. 3A). In reference to pre-, the cumulated distance covered was significantly increased at post- (pooled data: +1.9% ± 2.9%; $P = 0.019$, $\eta_p^2 = 0.02$), but without significant differences ($P = 0.149$, $\eta_p^2 = 0.11$) between RSH (+2.5% ± 2.6%) and RSN (+1.2% ± 3.3%) (Fig. 3D). A significant condition–time interaction was evident for V_{max} ($P = 0.035$, $\eta_p^2 = 0.01$), with higher values reached from pre- to post- for RSH (+2.2% ± 2.2%; $P = 0.004$, $\eta_p^2 = 0.04$) versus RSN (−0.1% ± 2.1%; $P = 0.908$, $\eta_p^2 = 0.00$) (Fig. 3B). In addition, V_{mean} increased ($P < 0.001$, $\eta_p^2 = 0.26$) from pre- to post- for RSH (+0.7% ± 2.0%) and RSN (+0.6% ± 2.0%), independent of the condition (Fig. 3C). No significant changes occurred for S_{dec} (Fig. 3E).

On-field aerobic capacity. No significant condition–time interaction ($P = 0.81$, $\eta_p^2 = 0.00$) or condition

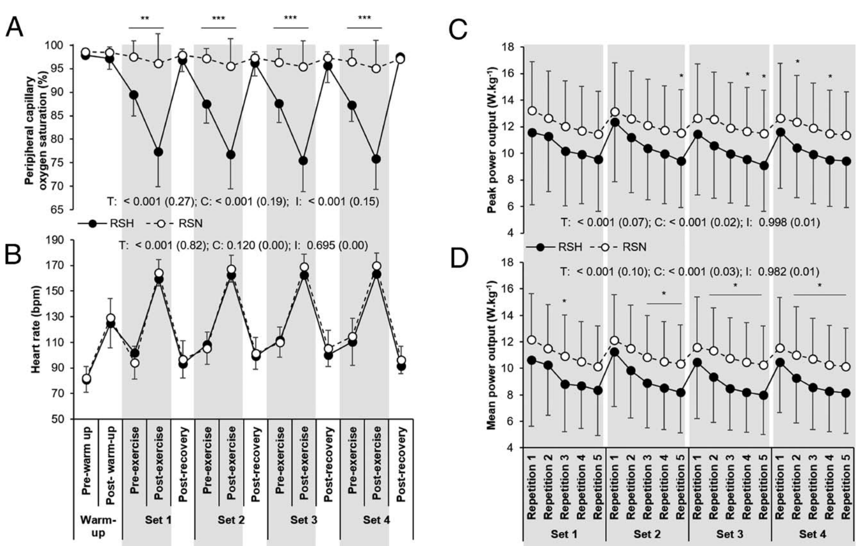

FIGURE 2—Peripheral capillary oxygen saturation (A), HR (B), and PPO and MPO (C and D, respectively) measured during the repeated-sprint training in hypoxia (RSH, *black circles* and *full line*) and in normoxia (RSN, *white circles* and *dotted line*). Values are the mean ± SD of the four repeated-sprint training sessions. Gray and white areas denote time spent inside and outside the normobaric hypoxic chamber, respectively. C, T, and I refer to ANOVA main effects of condition, time, and condition–time interaction, respectively, with *P* value and partial eta-squared (η_p^2) in brackets. ***P* < 0.01 and ****P* < 0.001 for *post hoc* comparison between conditions.

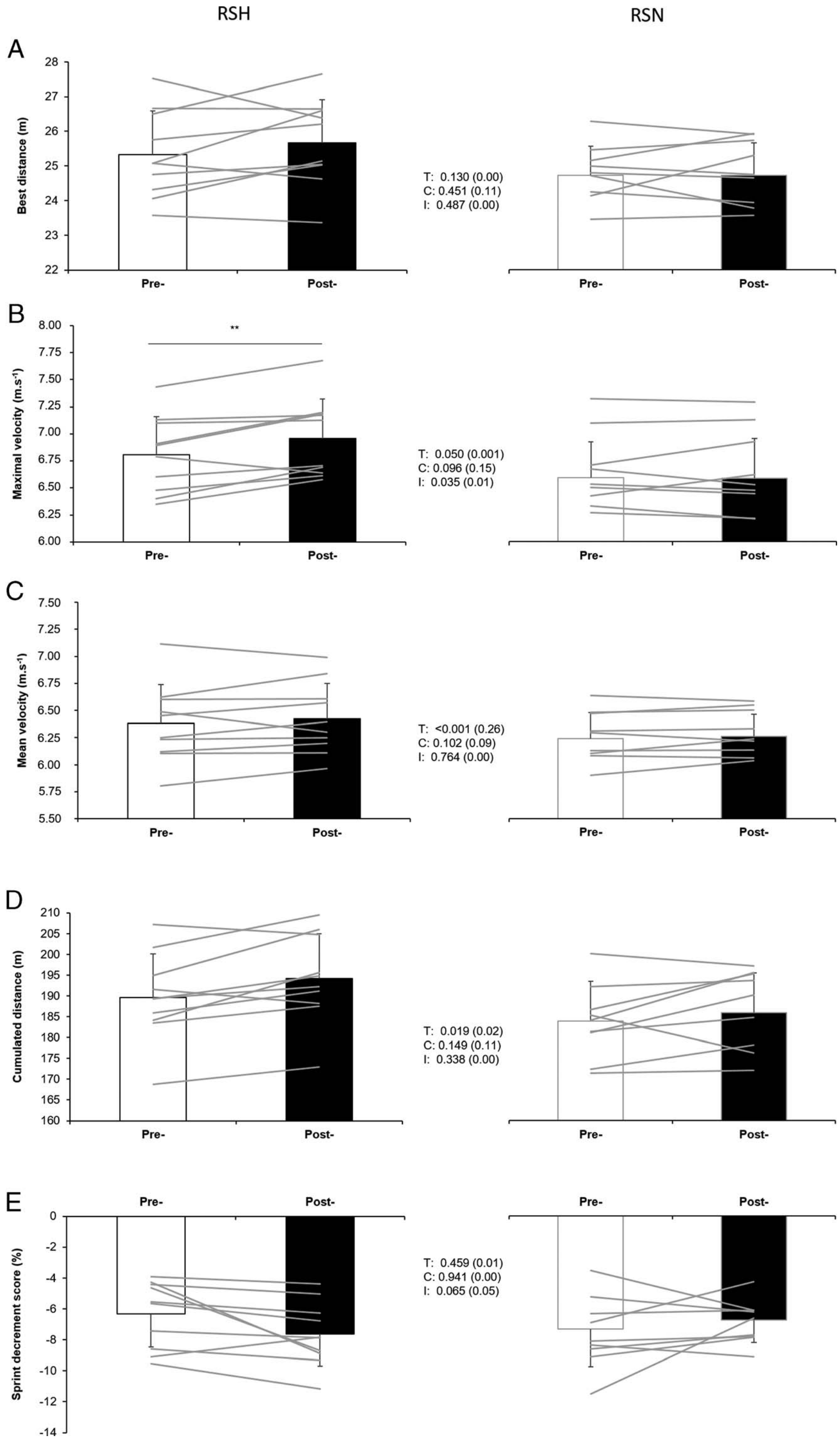

FIGURE 3—The best distance covered (A), maximal velocity (B), mean velocity (C), the cumulated distance covered (D), and sprint decrement score (E) of the treadmill repeated-sprint ability test performed before (pre-) and after (post-) repeated-sprint training in hypoxia (RSH) or in normoxia (RSN). Values are mean ± SD. C, T, and I refer to ANOVA main effects of condition, time, and condition–time interaction, respectively, with *P* value and partial eta-squared (η_p^2) in brackets. ***P* < 0.01 for difference between pre- and posttests.

(P = 0.45, η_p^2 = 0.03) and time main effects (P = 0.53, η_p^2 = 0.00) were found for the YYIR2 distance covered (RSH: 476 ± 127 vs 467 ± 108 m from pre- to post-, −0.6% ± 10.8%; RSN: 436 ± 106 vs 432 ± 86 m from pre- to post-, −1.6% ± 8.0%).

Brachial Artery Endothelial Function

Table 2 displays the pre- and postexperiment changes for the FMD-related variables. There was no significant condition–time interaction (all P > 0.240, all η_p^2 < 0.02) or

TABLE 2. Pre- and postexperiment FMD changes for baseline/peak arterial diameter and shear rates.

	RSH (n = 12)		RSN (n = 11)		ANOVA Main Effects (η_p^2)
	Pre-	Post-	Pre-	Post-	
Baseline diameter (mm)	3.45 ± 0.27	3.58 ± 0.31	3.70 ± 0.33	3.76 ± 0.32	T = 0.054 (0.02) C = 0.101 (0.11) I = 0.472 (0.00)
Peak diameter (mm)	3.69 ± 0.25	3.84 ± 0.29	3.98 ± 0.36	4.05 ± 0.38	T = 0.026 (0.02) C = 0.074 (0.14) I = 0.402 (0.00)
FMD change (%)	7.18 ± 2.63	7.23 ± 3.38	7.73 ± 1.90	7.65 ± 2.55	T = 0.988 (0.00) C = 0.609 (0.01) I = 0.932 (0.00)
Baseline shear rate (s^{-1})	139.64 ± 44.00	116.23 ± 32.22	151.51 ± 46.03	115.17 ± 41.51	T = 0.014 (0.09) C = 0.707 (0.00) I = 0.566 (0.00)
Peak shear rate (s^{-1})	749.6 ± 156.8	665.7 ± 208.4	764.1 ± 211.4	605.4 ± 223.4	T = 0.019 (0.07) C = 0.761 (0.00) I = 0.439 (0.01)
SR_{AUC} (s^{-1})	20,544.0 ± 6366.7	19,925.1 ± 9690.4	22,742.1 ± 7659.6	17,442.0 ± 7651.5	T = 0.230 (0.02) C = 0.957 (0.00) I = 0.337 (0.02)
Shear rate (s^{-1})	4726 ± 1429	3845 ± 2097	7278 ± 6852	3772 ± 1323	T = 0.057 (0.06) C = 0.300 (0.02) I = 0.240 (0.02)

Values are presented as mean ± SD.

condition effect (all $P > 0.074$, all $\eta_p^2 < 0.14$) for any brachial artery endothelial function variable. Only peak artery diameter (RSH, +3.9% ± 5.7%, and RSN, +1.7% ± 5.0%; $P = 0.026$, $\eta_p^2 = 0.02$), baseline shear rate (RSH, −10.1% ± 37.1%, and RSN, −18.7% ± 33.1%; $P = 0.014$, $\eta_p^2 = 0.09$), and peak shear rate (RSH, −7.7% ± 33.3%, and RSN, −19.4% ± 26.2%; $P = 0.019$, $\eta_p^2 = 0.07$) significantly changed over time.

DISCUSSION

This study examined a novel RSH protocol, intended to maximize the hypoxic stimulus while preserving training quality, on training/physiological adaptations and physical fitness (i.e., repeated-sprint ability and aerobic capacity) in world-class female rugby sevens players. The main findings were that a 9-d intervention (including four RSH sessions) using severe hypoxia exacerbated the physiological demand but lowered the external workload (greater reduction in mechanical power outputs) despite players observing normoxic interset rest periods in comparison to RSN. After the camp, similar improvements (except for V_{max}, which was significantly higher from pre- to post- in RSH) in repeated-sprint ability and brachial endothelial function occurred in the two groups. Our claim that hypoxic stress (~5000 m during exercise only) greater than that used in previous RSH studies (typically ~3000 m) provides putative effects in reference to the normoxic equivalent protocol is not supported.

Training response

Our novel RSH protocol used a severe hypoxic stimulus (FiO_2 of 10.6% equivalent to ~5000-m simulated altitude) during sets of sprints interspersed with normoxic (FiO_2 of 20.9%) between-sets recovery. Our intention was to maximize the hypoxic stimulus while preserving training quality, with the assumption to provide additional activation of anaerobic and neuromuscular pathways beyond that observed with "traditional" RSH routines (i.e., continuous exposure to moderate hypoxia throughout the session) (1,2,7–12). The hypoxic-induced physiological stimulus (end-exercise SpO_2 of ~76.8%) was larger than previously reported for repeated electromagnetically braked cycling sprints performed under moderate hypoxia levels (92.5% at FiO_2 = 14.5%, ~3000 m) (15). In addition, the decrease in mechanical power output was larger than we had envisioned, especially with the introduction of normoxic interset resting periods to facilitate the maintenance of PPO and MPO during the first repetition of subsequent set. However, this was insufficient to prevent a further decline across repetitions within each set in RSH, resulting in a lower external workload, despite an equivalent relative intensity (i.e., RPE), in RSH versus RSN.

Exercise performance tests

A 9-d precompetition training camp including four repeated-sprint sessions improved the treadmill repeated-sprint cumulated distance covered, although adding severe hypoxia did not induce larger performance gains. This finding aligns with the repeated-sprint ability-based shock microcycle (i.e., four to six sessions over 2 to 3 wk, three to four sets of 5 to 8 × 5—10-s sprints with 20- to 25-s passive/active recovery and 3- to 5-min rest) where comparable performance gains with or without hypoxic stress were reported (7,14,35,36). These results are practically relevant in scenarios where congested travel and competition schedules may interfere with traditional periodized training. Noteworthy, improvements in repeated-sprint ability (albeit statistical significance was only reached for V_{max}) were greater for RSH than RSN. These gains, even marginal, are important considering the prevalence of power-related factors (i.e., mid-thigh muscle cross-sectional area, adipose index, straight-line 20 m time, maximal sprint velocity, and countermovement jump with free arms) (37) and between-sets phosphocreatine resynthesis rate (38) in determining repeated-sprint

APPLIED SCIENCES

ability. Overall, this may confer a small advantage for RSH (likely due to the between-set resting periods in normoxia), especially in elite or world-class sport. However, the magnitude of the changes (i.e., between-group difference: 1.3%, 2.3%, and 0.2% for cumulated distance, $V_{\max}$, and V_{mean}, respectively) is lower compared with previous RSH intervention (i.e., 6.2% and 6.4% for PPO and MPO, respectively) performed at moderate hypoxia ($3 \times 8 \times 10$-s sprints, 20-s passive recovery at FiO_2 of 14.5% or 3000 m, four sessions over 2 wk) in elite male rugby union players (7). Altogether, this indicates that "*higher may not be better"* (19), as recently confirmed in swimmers (39), implying that additional RSH research on defining the optimal dose–response (including manipulating the exercise structure or using other ergogenic aid such as hyperoxia) is needed.

After RSH, divergent results have been reported on sport-specific endurance (8,13,14). In this study, the unchanged YYIR2 performance may indicate that the hypoxic dose could have been too stressful and elicited an exaggerated stimulus (i.e., exacerbated fatigue development eventually leading to overreaching) when measured immediately after the camp. This highlights the need to measure the delayed effects of any intervention (e.g., [35]) for a better interpretation of any individual change.

A possible explanation for the difference in repeated-sprint ability and on-field aerobic capacity, compared with previous studies of RSH in males (1,2,7–12), could relate to growing evidence that suggests sexual dimorphisms of gene expression between genders (40). This, in turn, might lead to differences in phenotypic expression, with considerable variances in muscle metabolic, contractile, and hemodynamic and perfusion properties (40–42). For instance, females typically display (i) higher repeated-sprint fatigue resistance (17), because of lower contractile dysfunction and metabolic disruption than males for an equivalent exercise dose (40), and (ii) lower sensitivity to hypoxic stimuli (16). Consequently, we expected (based on our empirical observations) higher resilience of tested female athletes using this novel RSH protocol. Compared with males, females exhibit faster peripheral and pulmonary oxygen extraction dynamics (41), greater vasodilatory responses (43), as well as higher density of capillaries per unit of skeletal muscle (42). This may possibly affect the oxygen cascade during RSH and, in turn, other RSH-induced physiological adaptations (e.g., mitochondrial biogenesis and angiogenesis) previously reported in males only (2,10).

Brachial artery endothelial function

Repetitive exercise-induced arterial shear stress (typical of short "all-out" repeated-cycling sprints) could stimulate endothelium-mediated arterial remodeling (arterial enlargement) (44). Although controversial (45), such endothelial functional and structural adaptations are deemed to occur beyond at least 4 wk of training (i.e., 15-min exercise, three times a week) (46). Here, brachial artery FMD was maintained after RSH and RSN, alongside reductions in shear rate patterns, presumably resulting from greater artery diameter (47). This may imply a beneficial response to shear stress-induced dilation after four sessions of repeated-cycling sprints over 9 d, irrespective of the condition. The lack of change in brachial artery FMD in both groups may relate to the inherent difficulty of upregulating arterial function deemed to be normal in this particular population (44). Alternatively, this could also result from faster training adaptations, thereafter superseded by structural adaptations (44), which in turn normalizes shear stress and negates the need for further functional enhancement. This last claim is perhaps particularly evident in RSN.

Peak arterial diameter increased significantly over time but without any condition (albeit being larger with additional hypoxic stress) or interaction effect. Because arterial lumen size and wall thickness (not measured here, susceptible to affect functional dilator response) both affect functional responsiveness (44), this may suggest an even more obvious compensatory structural remodeling in RSH versus RSN. Pending confirmatory results, these findings tend to align well with the recent conclusion by Lavier et al. (26) who used an animal model. Authors reported that, compared with RSN, RSH further increases nitric oxide bioavailability and improves endothelium-dependent vasorelaxation. In human, RSH is also known to upregulate the endothelial nitric oxide synthase pathway (10). Such endothelial-derived mechanism is particularly interesting as hypoxic stress may counteract the increase in circulating reactive oxygen species (which also mediate a reduction in nitric oxide bioavailability) generally reported in high-intensity exercise performed in normoxia (45). Collectively, the present findings (i) support the current statement that shear stress is key in any endothelium-dependent arterial remodeling process occurring from repeated-sprint training, and (ii) indicate that hypoxic training-induced compensatory vasodilatation and vascular mechanisms (among other adaptations) likely explain the possible putative effect of RSH versus RSN.

Limitations

Several limitations deserve attention. First, one may assume that brachial endothelial function cannot be extrapolated to active lower limb arteries as limb-specific differences have been reported with FMD (48). FMD-related assessment of brachial artery vasomotor function remains an accepted noninvasive surrogate for coronary artery vasomotor function (44), in particular, in special populations such as world-class athletes. That said, a more comprehensive understanding of arterial adaptations could be achieved by adopting multiple FMD position and time points. Second, both groups underwent repeated cold water immersion as part of their recovery process. Although we believe that similar benefits in restoring neuromuscular function during the 24- to 48-h recovery phase were achieved without affecting long-term muscle mitochondrial biogenesis and oxidative metabolism responses (49) after maximal-intensity sprint training, the potential different impact of RSH or RSN on vascular remodeling (50) remains uncertain. Third, although subjects' menstrual cycle was confidentially controlled by the team medical doctor, we were not able to consider oral contraceptive users susceptible to report blunted physiological adaptations compared with eumenorrhoeic females (51). However, it is possibly unlikely that subjects in various phases of their menstrual cycles have different fatigue tolerance responses (40). Possible fluctuation in estrogen (or progesterone) throughout the menstrual cycle that is susceptible to influence the vascular endothelial function (52), partly due to estrogen-associated increases in the nitric oxide bioavailability (53), could not be ruled out. Finally, although involving world-class athletes in research protocols is necessary, additional efforts should be made to consider a "gold-standard" randomized, double-blind, placebo-controlled, crossover study to disentangle the separated or synergistic effects of any intervention.

CONCLUSIONS

This study investigated novel repeated-sprint training in hypoxia protocol involving severe hypoxic stimuli (FiO_2 of 10.6% equivalent to ~5000 m simulated altitude) during sets of sprints interspersed with normoxic interset rest in world-class female rugby sevens players in a realistic setting (i.e., 9-d training camp before international competition). Adding only four cycle ergometer-based repeated-sprint sessions before a major international competition improved treadmill repeated-sprint ability and several vascular function indices, irrespective of the condition. Whether the augmented physiological stimulus, yet lowered mechanical output (despite normoxic interset rest), induced by a severe hypoxic stress adds putative effect remains unclear. Future studies should refine RSH best practice by determining the best combination between exercise structure and the hypoxic dose, with also direct comparison between females and males.

No potential conflict of interest was reported by the authors. This experiment was partly supported by Aspetar and by the French Rugby Union Federation. Data were collected using an instrumented treadmill funded by a QNRF grant (NPRP 4-760-3-217). The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine.

REFERENCES

1. Brocherie F, Girard O, Faiss R, Millet GP. Effects of repeated-sprint training in hypoxia on sea-level performance: a meta-analysis. *Sports Med*. 2017;47(8):1651–60.
2. Faiss R, Leger B, Vesin JM, et al. Significant molecular and systemic adaptations after repeated sprint training in hypoxia. *PLoS One*. 2013;8(2):e56522.
3. Girard O, Brocherie F, Millet GP. Effects of altitude/hypoxia on single- and multiple-sprint performance: a comprehensive review. *Sports Med*. 2017;47(10):1931–49.
4. Millet GP, Girard O, Beard AJ, Brocherie F. Repeated sprint training in hypoxia—an innovative method. *Dtsch Z Sportmed*. 2019;70:115–22.
5. Montero D, Lundby C. Repeated sprint training in hypoxia versus normoxia does not improve performance: a double-blind and cross-over study. *Int J Sports Physiol Perform*. 2017;12(2):161–7.
6. Goods PS, Dawson B, Landers GJ, Gore CJ, Peeling P. No additional benefit of repeat-sprint training in hypoxia than in normoxia on sea-level repeat-sprint ability. *J Sports Sci Med*. 2015;14(3):681–8.
7. Beard A, Ashby J, Chambers R, Brocherie F, Millet GP. Repeated-sprint training in hypoxia in international rugby union players. *Int J Sports Physiol Perform*. 2019;14(6):850–4.
8. Brocherie F, Girard O, Faiss R, Millet GP. High-intensity intermittent training in hypoxia: a double-blinded, placebo-controlled field study in youth football players. *J Strength Cond Res*. 2015;29(1):226–37.
9. Brechbuhl C, Brocherie F, Millet GP, Schmitt L. Effects of repeated-sprint training in hypoxia on tennis-specific performance in well-trained players. *Sports Med Int Open*. 2018;2(5):E123–32.
10. Brocherie F, Millet GP, D'Hulst G, Van Thienen R, Deldicque L, Girard O. Repeated maximal-intensity hypoxic exercise superimposed to hypoxic residence boosts skeletal muscle transcriptional responses in elite team-sport athletes. *Acta Physiol (Oxf)*. 2018;222(1):e12851.
11. Brechbuhl C, Brocherie F, Willis SJ, et al. On the use of the repeated-sprint training in hypoxia in tennis. *Front Physiol*. 2020; 11:588821.
12. Faiss R, Willis S, Born DP, et al. Repeated double-poling sprint training in hypoxia by competitive cross-country skiers. *Med Sci Sports Exerc*. 2015;47(4):809–17.
13. Galvin HM, Cooke K, Sumners DP, Mileva KN, Bowtell JL. Repeated sprint training in normobaric hypoxia. *Br J Sports Med*. 2013; 47(1 Suppl):i74–9.
14. Hamlin MJ, Olsen PD, Marshall HC, Lizamore CA, Elliot CA. Hypoxic repeat sprint training improves rugby player's repeated sprint but not endurance performance. *Front Physiol*. 2017;8:24.
15. Kasai N, Mizuno S, Ishimoto S, Sakamoto E, Maruta M, Goto K. Effect of training in hypoxia on repeated sprint performance in female athletes. *Springerplus*. 2015;4:310.
16. Billaut F, Smith K. Sex alters impact of repeated bouts of sprint exercise on neuromuscular activity in trained athletes. *Appl Physiol Nutr Metab*. 2009;34(4):689–99.
17. Billaut F, Bishop D. Muscle fatigue in males and females during multiple-sprint exercise. *Sports Med*. 2009;39(4):257–78.
18. Bowtell JL, Cooke K, Turner R, Mileva KN, Sumners DP. Acute physiological and performance responses to repeated sprints in varying degrees of hypoxia. *J Sci Med Sport*. 2014;17(4):399–403.
19. Goods PSR, Dawson BT, Landers GJ, Gore CJ, Peeling P. Effect of different simulated altitudes on repeat-sprint performance in team-sport athletes. *Int J Sports Physiol Perform*. 2014;9(5):857–62.
20. Calbet JAL, De Paz JA, Garatachea N, Cabeza de Vaca S, Chavarren J. Anaerobic energy provision does not limit Wingate exercise performance in endurance-trained cyclists. *J Appl Physiol (1985)*. 2003;94(2):668–76.
21. Weyand PG, Lee CS, Martinez-Ruiz R, Bundle MW, Bellizzi MJ, Wright S. High-speed running performance is largely unaffected by hypoxic reductions in aerobic power. *J Appl Physiol (1985)*. 1999; 86(6):2059–64.
22. Brocherie F, Millet GP, Morin JB, Girard O. Mechanical alterations to repeated treadmill sprints in normobaric hypoxia. *Med Sci Sports Exerc*. 2016;48(8):1570–9.
23. Townsend N, Brocherie F, Millet GP, Girard O. Central and peripheral muscle fatigue following repeated-sprint running in moderate and severe hypoxia. *Exp Physiol*. 2021;106(1):126–38.
24. Reboul C, Tanguy S, Dauzat M, Obert P. Altitude negates the benefits of aerobic training on the vascular adaptations in rats. *Med Sci Sports Exerc*. 2005;37(6):979–85.
25. Casey DP, Joyner MJ. Compensatory vasodilatation during hypoxic exercise: mechanisms responsible for matching oxygen supply to demand. *J Physiol*. 2012;590(24):6321–6.
26. Lavier J, Beaumann M, Menetrey S, et al. High-intensity exercise in hypoxia improves endothelial function via increased nitric oxide bioavailability in C57BL/6 mice. *Acta Physiol (Oxf)*. 2021;233(2):e13700.
27. Thijssen DH, Black MA, Pyke KE, et al. Assessment of flow-mediated dilation in humans: a methodological and physiological guideline. *Am J Physiol Heart Circ Physiol*. 2011;300(1):H2–12.
28. Brocherie F, Millet GP, Girard O. Psychophysiological responses to repeated-sprint training in normobaric hypoxia and normoxia. *Int J Sports Physiol Perform*. 2017;12(1):115–23.

29. Couderc A, Thomas C, Lacome M, et al. Movement patterns and metabolic responses during an international rugby sevens tournament. *Int J Sports Physiol Perform*. 2017;12(7):901–7.
30. Girard O, Brocherie F, Morin JB, Millet GP. Mechanical alterations during interval-training treadmill runs in high-level male team-sport players. *J Sci Med Sport*. 2017;20(1):87–91.
31. Krustrup P, Mohr M, Nybo L, Jensen JM, Nielsen JJ, Bangsbo J. The Yo-Yo IR2 test: physiological response, reliability, and application to elite soccer. *Med Sci Sports Exerc*. 2006;38(9):1666–73.
32. Black MA, Cable NT, Thijssen DHJ, Green DJ. Importance of measuring the time course of flow-mediated dilatation in humans. *Hypertension*. 2008;51(2):203–10.
33. Bailey TG, Birk GK, Cable NT, et al. Remote ischemic preconditioning prevents reduction in brachial artery flow-mediated dilation after strenuous exercise. *Am J Physiol Heart Circ Physiol*. 2012;303(5):H533–8.
34. Atkinson G, Batterham AM. The percentage flow-mediated dilation index: a large-sample investigation of its appropriateness, potential for bias and causal nexus in vascular medicine. *Vasc Med*. 2013;18(6):354–65.
35. Brechbuhl C, Schmitt L, Millet GP, Brocherie F. Shock microcycle of repeated-sprint training in hypoxia and tennis performance: case study in a rookie professional player. *Int J Sports Sci Coach*. 2018;13(5):723–8.
36. Brocherie F, Perez J, Guilhem G. Effects of a 14-day high-intensity shock microcycle in high-level ice hockey players' fitness. *J Strength Cond Res*. 2022;36(8):2247–52.
37. Brocherie F, Girard O, Forchino F, Al Haddad H, Dos Santos GA, Millet GP. Relationships between anthropometric measures and athletic performance, with special reference to repeated-sprint ability, in the Qatar national soccer team. *J Sports Sci*. 2014;32(13):1243–54.
38. Mendez-Villanueva A, Edge J, Suriano R, Hamer P, Bishop D. The recovery of repeated-sprint exercise is associated with PCr resynthesis, while muscle pH and EMG amplitude remain depressed. *PLoS One*. 2012;7(12):e51977.
39. Camacho-Cardenosa M, Camacho-Cardenosa A, Gonzalez-Custodio A, Zapata V, Olcina G. Effects of swimming-specific repeated-sprint training in hypoxia training in swimmers. *Front Sports Act Living*. 2020;2:100.
40. Ansdell P, Thomas K, Hicks KM, Hunter SK, Howatson G, Goodall S. Physiological sex differences affect the integrative response to exercise: acute and chronic implications. *Exp Physiol*. 2020;105(12):2007–21.
41. Beltrame T, Villar R, Hughson RL. Sex differences in the oxygen delivery, extraction, and uptake during moderate-walking exercise transition. *Appl Physiol Nutr Metab*. 2017;42(9):994–1000.
42. Roepstorff C, Thiele M, Hillig T, et al. Higher skeletal muscle alpha2AMPK activation and lower energy charge and fat oxidation in men than in women during submaximal exercise. *J Physiol*. 2006;574(Pt 1):125–38.
43. Parker BA, Smithmyer SL, Pelberg JA, Mishkin AD, Herr MD, Proctor DN. Sex differences in leg vasodilation during graded knee extensor exercise in young adults. *J Appl Physiol (1985)*. 2007;103(5):1583–91.
44. Green DJ, Spence A, Rowley N, Thijssen DH, Naylor LH. Vascular adaptation in athletes: is there an 'athlete's artery'? *Exp Physiol*. 2012;97(3):295–304.
45. Bergholm R, Makimattila S, Valkonen M, et al. Intense physical training decreases circulating antioxidants and endothelium-dependent vasodilatation in vivo. *Atherosclerosis*. 1999;145(2):341–9.
46. Tinken TM, Thijssen DH, Black MA, Cable NT, Green DJ. Time course of change in vasodilator function and capacity in response to exercise training in humans. *J Physiol*. 2008;586(20):5003–12.
47. Madsen SM, Thorup AC, Overgaard K, Bjerre M, Jeppesen PB. Functional and structural vascular adaptations following 8 weeks of low volume high intensity interval training in lower leg of type 2 diabetes patients and individuals at high risk of metabolic syndrome. *Arch Physiol Biochem*. 2015;121(5):178–86.
48. Thijssen DH, Rowley N, Padilla J, et al. Relationship between upper and lower limb conduit artery vasodilator function in humans. *J Appl Physiol (1985)*. 2011;111(1):244–50.
49. Chaillou T, Treigyte V, Mosely S, Brazaitis M, Venckunas T, Cheng AJ. Functional impact of post-exercise cooling and heating on recovery and training adaptations: application to resistance, endurance, and sprint exercise. *Sports Med Open*. 2022;8(1):37.
50. Yamane M, Ohnishi N, Matsumoto T. Does regular post-exercise cold application attenuate trained muscle adaptation? *Int J Sports Med*. 2015;36(8):647–53.
51. Sims ST, Heather AK. Myths and methodologies: reducing scientific design ambiguity in studies comparing sexes and/or menstrual cycle phases. *Exp Physiol*. 2018;103(10):1309–17.
52. Williams MR, Westerman RA, Kingwell BA, et al. Variations in endothelial function and arterial compliance during the menstrual cycle. *J Clin Endocrinol Metab*. 2001;86(11):5389–95.
53. Mendelsohn ME. Genomic and nongenomic effects of estrogen in the vasculature. *Am J Cardiol*. 2002;90(1A):3F–6F.

APPLIED SCIENCES